

VISHWARAJ SURTHI

Full Stack Developer • React • Node.js/C# • Python • System Design • AWS

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PROFESSIONAL SUMMARY

Full Stack Developer with 5+ years shipping production platforms across the stack, from responsive React/Next.js front ends to backend services, APIs, and relational/distributed data stores. Skew toward backend architecture, system design, and code optimization, with a track record of solving complex data-processing and resource-optimization problems using strong algorithmic and data-structure fundamentals. Effective across multiple languages and ecosystems (TypeScript/JavaScript, Java, Python), choosing the right tool for the problem rather than a single stack. Experienced building well-designed APIs that integrate products into a unified architecture, owning services end to end in production (testing, observability, performance, on-call), and collaborating directly with product leads on scoping, estimation, and technical decisions. Hands-on with AWS, containers, CI/CD, and dashboard/Bi-style, high-throughput data-driven products.

CORE SKILLS

Languages & Runtimes: TypeScript, JavaScript (ES2024), Python, C#, .Net, ASP.NET Core (.NET 8), Node.js

Front-End: React 18, Next.js (SSR/SSG), Redux Toolkit, React Query, Apollo Client, HTML5, CSS3

Back-End & APIs: Spring Boot, Express.js, REST, GraphQL, SOAP/WSDL, microservices, API design, OAuth 2.0 / OIDC, JWT

Data & Messaging: MySQL, PostgreSQL, MongoDB, Redis, Kafka, WebSockets, data modeling

Algorithms & System Design: Data structures, distributed systems, caching strategies, query optimization, resource optimization, scalability

Cloud & DevOps: AWS (Lambda, EKS, S3, CloudFront, CloudWatch, API Gateway), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins)

Testing & Observability: Jest, JUnit, React Testing Library, Playwright, Cypress, Datadog, on-call/incident response

PROFESSIONAL EXPERIENCE

Matrix Medical Network — *Phoenix, AZ (Remote)*

Oct 2025 – Present

Full Stack Developer

- Developed new features and enhanced existing capabilities for an **Intelligent Care Management Platform (ICMP)** that helped clinicians review patient history, care gap identification, and generate AI-assisted patient summaries during in-home and virtual assessments. Built **Next.js**, **React 18/TypeScript** user interfaces and **Node.js (NestJS)/ ASP.NET Core (.NET 8) microservices**, taking ownership of API design, data modeling, and end-to-end healthcare capabilities.
- Designed and developed clinical decision-support microservices using **C#/.NET Core** and **REST APIs** with **OpenAPI/Swagger**, separating eligibility rules, risk calculations, and workflow logic from the frontend application. This improved service maintainability, **reduced duplicated business logic**, and allowed clinical capabilities to be reused across multiple applications.
- Integrated **Azure OpenAI services** to build AI-assisted patient summarization with **Azure AI Search** and clinical recommendation capabilities using Retrieval-Augmented Generation (**RAG**). Developed backend orchestration services that combined patient records, assessment results, and approved clinical/healthcare guidelines to generate context-aware recommendations while keeping clinicians/human in control of final decisions.
- Built configurable workflow services for clinical assessments and care pathways, allowing healthcare teams to introduce new programs, update decision rules, and modify assessment flows without requiring major application releases. Designed schema-driven APIs consumed by **Next.js** applications to support flexible clinical workflows across different care models.
- Improved the performance of patient and clinician services by optimizing **MongoDB**, **PostgreSQL** and **MySQL database schemas**, tuning complex SQL queries, refining indexing strategies, and introducing **Redis caching** for frequently accessed data. These changes improved API responsiveness for high-volume clinical searches and assessment workflows.
- Strengthened platform security by implementing **OAuth 2.0**, **OpenID Connect**, **JWT-based authentication**, **Azure Entra ID** and role-based access control (**RBAC**) across distributed microservices. Worked closely with the identity team to enforce HIPAA-compliant, **Zero Trust security practices** and protect sensitive patient information throughout the platform.
- Containerized backend services using **Docker** and deployed applications through **Azure Container Apps** with automated CI/CD pipelines built using **GitHub Actions** and **Jenkins**. Established deployment workflows across development, QA, and production environments while using **Azure Monitor** and CloudWatch-style observability practices for application health tracking, logging, and alerting.

- Improved production reliability by implementing structured logging, monitoring dashboards, and automated health checks across application services. Participated in incident troubleshooting, performance analysis, and production support activities to maintain availability of critical clinician workflows.
- Worked closely with product managers, architects, QA engineers, and cross-functional teams throughout the software development lifecycle, participating in technical design discussions, code reviews, and production issue resolution.

BMO US — *Chicago, IL*

Aug 2023 – Sep 2025

Full Stack Engineer — Technology & Operations

- Built core services and modern **Next.js/React** applications for a cloud-native digital banking and payment platform supporting customer onboarding, account servicing, and high-volume payment workflows. Built **Python (FastAPI)** and **ASP.NET Core (.NET)** services, contributing across API design, service architecture, and end-to-end feature delivery.
- Designed microservices for separating payment processing, account management, and notifications using **Python (FastAPI)** and **ASP.NET Core**, using clean controller/service/repository separation, simplifying boundaries and enabling each service to scale and deploy independently. Published versioned **REST APIs** with **OpenAPI/Swagger**, simplifying service integration and long-term maintainability.
- Implemented event-driven transaction processing using **Apache Kafka**, allowing payment, transfer, notification, and audit services to communicate asynchronously while improving fault tolerance, scalability, and transaction throughput during peak banking operations.
- Engineered idempotent payment processing and retry-safe transaction workflows within **ASP.NET Core** services, preventing duplicate ledger updates during retries, network interruptions, and service restarts while improving reconciliation across distributed systems.
- Developed reusable **Next.js** and React components using **TypeScript**, **Redux Toolkit**, **TanStack Query**, and Material UI for onboarding, account servicing, and payment workflows. Worked with backend teams to integrate secure APIs and implement **OAuth 2.0**, **JWT**, and role-based authorization across customer and internal banking applications.
- Deployed and scaled services across multiple AWS environments using containerized deployments (**Docker/Kubernetes**) and automated CI/CD pipelines (**Jenkins/GitHub Actions**), enabling consistent, repeatable releases from dev through production.
- Tuned **PostgreSQL/MySQL queries** and indexing for high-frequency onboarding and transaction endpoints, cutting the rate of slow calls and improving p95 latency under peak load.
- Improved production reliability by implementing observability with **CloudWatch**, **Datadog**, and **OpenTelemetry**, creating dashboards, distributed tracing, and proactive alerting that accelerated issue detection and reduced recovery time for business-critical payment services.
- Collaborated directly with product leads and stakeholders on scoping, estimation, and technical decisions within the pod, and mentored engineers through code review on backend service design and API contracts.

GlobalLogic — *Hyderabad, India*

Jan 2020 – Jun 2022

Software Analyst

- Built a full-stack patient-facing scheduling and appointment portal — **Angular 8** front end on **Java Spring Boot** backend services and APIs — migrating a legacy **AngularJS** codebase in a phased rollout that reduced user error rates by **20%** and improved first-contentful-paint by 1.2s.
- Designed and implemented REST and SOAP/WSDL Java **Spring Boot APIs** parsing XSD-defined XML schemas for appointment validation and billing, applying algorithmic optimization to availability-matching logic to keep interactions responsive under load.
- Improved **MongoDB**, **MySQL** query performance by **50%** through index design, query refactoring, and data-model changes, eliminating timeout errors that had been surfacing in the patient-search service under peak load.
- Built accessible booking and intake forms in **JavaScript/Angular** with inline validation backed by validation rules enforced in the Spring Boot service layer, reducing average form completion time by 25%.
- Implemented Git-based CI/CD workflows (**Jenkins**) spanning front-end and backend, accelerating release cadence from bi-weekly to weekly with a reliable automated test gate before every merge.
- Integrated **Redis pub/sub** messaging on the backend for real-time appointment notifications, delivering confirmation alerts to clinicians within 200ms without a page reload.
- Participated in technical interviews for full-stack engineering candidates, evaluating system design, algorithms, and API design fundamentals, and contributed hiring recommendations.

EDUCATION

Indiana Institute of Technology Aug 2023 – Dec 2024

Master of Science, Information Systems — Fort Wayne, IN